

Beyond Single-Answer RAG: Viewpoint Coverage Retrieval for Financial Decision Support

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Abstract

Retrieval-augmented generation (RAG) systems are typically optimized to retrieve relevant evidence and synthesize a single coherent answer. However, many decision-oriented information needs do not have a single universally correct answer. In domains such as personal finance, multiple recommendations may be reasonable depending on assumptions about interest rates, liquidity, time horizon, tax treatment, market conditions, and risk tolerance. A single synthesized answer can therefore obscure genuine disagreement, caveats, and answer-changing conditions. We introduce EPIC, an epistemic retrieval pipeline for financial decision support that retrieves candidate documents, decomposes them into structured argument units, clusters claims using a direction-aware perspective discovery procedure, and presents grounded perspective panels rather than a final recommendation. We evaluate EPIC v4 across five financial decision queries and four retrieval depths ($K = 5, 10, 15, 20$), comparing it against random grouping, relevance-order grouping, semantic-only clustering, direction-only grouping, and unclear-handling ablations. The results show that retrieval depth strongly shapes the epistemic landscape: very small retrieval pools often miss major viewpoint families, while mid-range settings, especially $K = 15$, frequently provide a better balance between coverage and redundancy. EPIC performs strongest on queries with naturally separable stance families, such as student-loan repayment versus investing and emergency funds versus credit-card debt, but is less successful when the source ecosystem poorly matches the exact decision framing. Semantic-only clustering remains a strong intrinsic baseline, especially on narrower or noisier queries. These findings suggest that EPIC is best understood not as a universally superior clusterer but as a schema-guided argument-discovery and viewpoint-induction pipeline for deliberation-facing RAG interfaces. Notably, when source coverage is thin, EPIC degrades gracefully by producing narrower but fully grounded outputs rather than inflating diversity with unsupported perspectives – a property that distinguishes it from generation-centered approaches.

Keywords

retrieval-augmented generation, information retrieval, viewpoint diversity, decision support, epistemic coverage, trust calibration, personal finance

1 Introduction

Modern retrieval systems are not merely tools for finding and accessing information; rather, they increasingly shape how people make decisions. A user asking “*Should I pay off student loans or start investing?*” is not merely seeking a fact. They are asking for assistance in navigating a trade-off. The appropriate answer depends on a number of extrinsic factors, including interest rates, emergency savings, employer retirement matching, and tax advantages, as well as intrinsic factors such as market expectations, risk tolerance, and personal goals. In such scenarios where the answer is subjective and is influenced by a plethora of associated conditions and caveats, different sources may make different but legitimate recommendations. A certified financial planner may emphasize risk and debt elimination; an investing-oriented article may emphasize long-run market returns; an online community may emphasize psychological comfort or the flexibility of being debt-free. Therefore, the goal is no longer just relevance, but coverage: surfacing the range of perspectives that shape a decision.

This shift is especially important because users increasingly turn to conversational language models such as *ChatGPT*, *Claude AI* and *Gemini* for tasks that traditional retrieval systems once supported. Unlike ranked search results, large language models provide more direct synthesis, conversational follow-ups, and a sense of personalization, making them even more attractive for complex decision-oriented queries. However, these systems may not actually be suitable for deliberative decision-making or for surfacing disagreements. Their responses are largely generated by parametric memory, which can reflect overrepresented viewpoints in the training data. They may also produce unsupported or hallucinated claims, a limitation that persists even in state-of-the-art models [4, 16, 19]. These are the exact reasons why simply prompting the LLM to “generate multiple viewpoints” risks producing a facade of epistemic diversity that lacks grounding and comprehensive representation of the evidence landscape. In addition, these systems often compress complex decision landscapes into a single, fluent response, making one framing of the problem appear more settled than it is. Recent work on epistemic diversity shows that nearly all tested LLMs produce less epistemically diverse outputs than basic web search, and that larger models tend to exhibit lower epistemic diversity, contributing to the risk of *knowledge collapse* [25]. Consequently, for such queries, the problem is not only whether an answer is correct or grounded, but also whether it preserves the range of disagreement, uncertainty, and conditionality needed for informed judgment.

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Modern RAG systems address part of this problem, but remain poorly matched to decision-oriented retrieval. A standard RAG pipeline retrieves top-ranked evidence and asks a language model to synthesize a coherent response. This improves factual grounding relative to purely parametric generation, but the final output is still usually a single narrative [18]. This narrative more often than not flattens disagreement, hides minority but valid viewpoints, and makes one recommendation appear more prominent than the underlying evidence supports. This phenomenon is known as *single-answer bias*. For decision-oriented queries, this is not simply a presentation issue. It can encourage passive reliance by giving users the impression that the system has *resolved* a tradeoff that is inherently conditional. Prior work on AI-assisted decision-making shows that users may over-rely on AI recommendations rather than critically evaluate them, motivating designs that increase people’s cognitive motivation for engaging analytically rather than passively accepting a system verdict [6].

These limitations suggest that decision-oriented retrieval should not be framed as a problem of producing a single best answer. Instead, the system should help users inspect the space of plausible answers and understand why different sources may reasonably disagree. This project, hence, studies a different retrieval objective: *viewpoint coverage retrieval*. Rather than asking a system to retrieve the best documents or produce the best answer, we ask whether it can retrieve and organize distinct grounded perspectives so that users can deliberate. We focus on *personal finance* because it is a domain where advice is consequential, source incentives vary, and recommendations are often conditional rather than universally correct.

We present EPIC, an epistemic RAG pipeline that operationalizes viewpoint coverage for financial decision queries. EPIC first retrieves candidate documents using BM25. It then decomposes each document into structured argument units consisting of a main claim, supporting text, conditions, caveats, reason types, and a query-relative decision direction. Instead of clustering full passages, EPIC clusters claim representations within direction buckets, preserving the distinction between options such as paying off student loans, investing, or doing both. Each resulting cluster is summarized in a perspective panel that includes a label, core summary, key reasons, source grounding, conditions, and caveats. The interface deliberately withholds a final recommendation.

This paper makes three contributions. First, we define viewpoint coverage retrieval as a decision-support retrieval problem distinct from standard relevance ranking and single-answer RAG. Second, we introduce EPIC, a lightly supervised pipeline that combines structured argument decomposition with direction-aware perspective discovery. Third, we conduct a multi-query evaluation of EPIC across five financial decision scenarios, four retrieval depths, and several ablations. We find that EPIC reliably outperforms weak baselines such as random grouping and relevance-order grouping, and is strongest when the corpus contains clearly separable stance families. However, semantic-only clustering remains competitive or stronger on some intrinsic geometric metrics. We interpret this not as a negative result to hide, but as a methodological finding: systems for deliberation-support retrieval should be evaluated not only by embedding-space separation, but also by directional coverage, perspective distinctness, and downstream decision usefulness.

2 Related Work

2.1 Search Result Diversification

Most traditional retrieval systems are built around relevance: given a query, they return a ranked list of documents that best match it. This is effective when the user’s primary goal is to find a specific document or collect relevant evidence (*high recall*). But with the modern ginormous online information spaces, the top results more often than not contain substantial redundancy, which may sometimes overrepresent the most common framing of an issue. For decision-oriented queries, the problem is therefore not only finding relevant documents but also ensuring coverage of distinct perspectives. Prior work in *result diversification* has addressed the problem of excessive redundancy in a ranked list of documents [7, 21]. Methods such as maximal marginal relevance (MMR) trade off relevance and novelty to reduce semantic redundancy. They incorporate semantic diversity into traditional relevance ranking by penalizing candidate documents that are similar to already selected results. However, it is important to note that *semantic novelty* is *not* the same as *epistemic diversity*. Two documents can be semantically similar while making opposite recommendations, and two documents can be semantically different while supporting the same decision direction. For example, two articles both discuss interest rates and investment returns, but one recommends paying off debt, while the other suggests investing. In contrast, an online community post may use a completely different language to discuss the psychological relief of a debt-free future while still supporting the debt-first claim. The primary goal of our pipeline is to build on the motivation for diversification while shifting the idea of diversity from *document similarity* to *decision-relevant viewpoint structure*.

2.2 Personal Finance as a Decision-Support Domain

With the rise of general-purpose AI systems such as *ChatGPT*, *Claude*, and *Gemini*, as well as financial assistants such as *Erica by Bank of America* and *Eno by Capital One*, individuals are increasingly turning to these systems for guidance on budgeting, investing, debt management, mortgage, and loan payments. This shift has led to a growing body of research examining whether such systems can provide reliable personal financial advice, especially in settings where recommendations depend on user goals, risk tolerance, and changing financial circumstances. Prior work has used finance as a domain for financial question answering, numerical reasoning, and RAG evaluation, including FinQA [8], ConvFinQA [9], FinanceBench [14], and FinDER [10]. More recent work has also started using personal finance as a domain for evaluating LLM-based advice. Schlosky et al. [22] assessed ChatGPT’s recommendations for 21 standalone personal finance cases to find that it may be a good starting point, but it gives generic advice and may overlook alternative solutions, viewpoints, and recommendation priorities. Choukhmane et al. [11] study AI-generated personal financial advice in dynamic life-cycle settings and emphasize that financial advice is difficult to evaluate because decisions heavily depend on uncertainty, preferences, and changing individual circumstances. These works, however, primarily focus on evaluating the quality, correctness, or usefulness of generated advice. In

contrast, our study treats personal finance as a retrieval problem: rather than asking whether an LLM can produce good advice, we ask whether a retrieval system can surface distinct, grounded perspectives that preserve disagreement, trade-offs, and conditions. This, in turn, could potentially improve the quality of judgment essential to better decision-making.

2.3 LLMs for Decision Support

Large language models, especially conversation-based LLMs, are increasingly being used as decision-support tools across several domains because they can synthesize and summarize information, answer follow-up questions, and present advice in a conversational form. These models, combined with easy-to-use interfaces, make them extremely attractive for tasks in which users must interpret complex information rather than simply retrieve a fact. However, it has been well-documented that LLMs can produce hallucinated or unsupported claims, even with state-of-the-art models [16, 19]. These systems, therefore, present a double-edged sword: although their responses may not always be grounded, their fluency can create a misleading impression of reliability. This is especially problematic in decision-oriented settings, where the central issue is often not only factual correctness but also whether the system preserves uncertainty, disagreement, and different viewpoints. Recent work on epistemic diversity further shows that LLM outputs can be less diverse than traditional web search, suggesting that language models may narrow the range of viewpoints users encounter rather than broaden it [25]. Our work builds on this concern by treating decision support not as a task of *synthesizing* better advice, but rather as a retrieval problem. We aim to create a pipeline that surfaces distinct, grounded perspectives for user deliberation.

2.4 Retrieval-Augmented Generation

The primary aim of retrieval-augmented generation is to combine document retrieval with LLM models to ground outputs in external evidence [18]. Standard RAG pipelines are usually designed around answer accuracy or factual grounding. They retrieve relevant documents, condition the generation on those documents, and synthesize a response. This architecture has been shown to be successful on many knowledge-intensive tasks, such as open-domain QA [15, 18]. Wright et al. [25] also found that RAG can improve epistemic diversity as compared to parametric-only LLM generation, but it still lags in comparison with traditional web search. In addition, this improvement is moderated by the diversity of the retrieval corpus itself (*corpus skew*). This directly motivates our focus on retrieval itself: if the retrieved evidence is narrow or if generation collapses retrieved evidence into a single narrative, RAG may still fail to support decision-oriented deliberation.

2.5 AI Overreliance and Trust Calibration

While RAG systems can help ground synthesized responses in facts, they compress uncertainty, disagreement, and diverse viewpoints into a single, fluent narrative. In decision-making contexts, this lack of *epistemic diversity* can perpetuate *passive reliance* or, at times, *over-reliance* on AI recommendations. Prior work suggests that interface interventions that engage users analytically through cognitive motivation can reduce over-reliance [6]. Our project connects

this concern to the fundamental design of retrieval. Rather than treating over-reliance simply as an interface-level problem, we ask whether retrieval systems can be designed to preserve disagreement and uncertainty before generation. We then use the interface to expose this structure to users through grounded perspective panels rather than a single system recommendation. Our long-term goal is to study whether multi-perspective retrieval combined with the interfaces improves trust calibration and perceived agency compared with standard search or single-answer RAG.

2.6 Argument Retrieval and Stance Organization

The primary goal of *argument retrieval* is to retrieve arguments relevant to controversial questions, often organizing them into pro and con positions [5]. This is closely related to our problem statement, but it is worth noting that decision-support advice is not always binary. Financial advice often involves hybrid positions, threshold rules, conditional recommendations, and caveats. For example, a source may recommend investing if student loan interest rates are low, paying off debt if interest rates are high, or splitting available funds between debt repayment and retirement contributions. EPIC, our pipeline, thus decomposes documents into claims, conditions, caveats, and reason types before clustering.

2.7 Diverse and Viewpoint-Oriented RAG

Recent work has started to explicitly incorporate diversity into RAG systems. DF-RAG [17] proposes a query-aware diversity mechanism for RAG, arguing that standard cosine-similarity retrieval often returns redundant context and limits information recall. The proposed system builds on Maximal Marginal Relevance (MMR) [7] to dynamically adjust the relevance-diversity trade-off at test time and improve performance on reasoning-intensive QA benchmarks. This work shows that diversity-aware retrieval can improve RAG performance, but its primary objective remains answer accuracy for QA rather than organizing the epistemic landscape for user deliberation.

DIVERGE [13] is much closer to our motivation because it studies open-ended information seeking, where multiple plausible answers may exist. It argues that standard RAG systems often underutilize retrieved context diversity and that simply increasing retrieval diversity does not necessarily produce "diverse generations". To address this, DIVERGE inspired by the *Plan-and-Solve* agentic designs [24], repeatedly prompts the model to look for viewpoints it has not yet covered and uses past generations as memory to expand the range of perspectives. However, DIVERGE remains primarily generation-centered: it aims to produce diverse responses or viewpoints, whereas our pipeline aims to construct an *auditable map of grounded perspectives* from retrieved evidence.

EPIC pipeline is complementary to these approaches but shifts the objective from *diverse answer generation* to *viewpoint coverage retrieval*. We do not rely on the model to generate diverse perspectives; rather, EPIC decomposes retrieved documents into claims, conditions, caveats, reason types, and decision directions, then clusters these units into grounded perspective panels. In this sense, EPIC treats diversity as a property of the organized evidence landscape, not merely as a property of the final generated text.

3 Problem Formulation & Research Question

3.1 Defining Decision-oriented Query

In this paper, we define a *decision-oriented query* as one in which the user must choose between two or more plausible actions. In addition to this, the *appropriate answer* depends on assumptions or context.

3.2 Problem Definition

Let q be a query with two focal options, a and b . A document collection D contains documents that may support option a , option b , a hybrid approach, or no clear recommendation.

Our goal is NOT to return the single highest-ranked document or to synthesize a single final answer. Rather, we aim to produce a set of perspective clusters $C = \{c_1, \dots, c_k\}$ such that each cluster is *internally coherent and mutually distinct* from other clusters, and collectively covers the major decision directions present in the corpus.

We call this task *viewpoint coverage retrieval*. A successful system should ideally satisfy these properties:

- (1) retrieve relevant evidence for the query;
- (2) identify the main decision direction of each argument;
- (3) preserve conditions and caveats rather than mixing them into the main claim representation; and
- (4) produces an interface-ready representation that allows users to compare perspectives without forcing a final verdict.

3.3 Research Question

Using this formulation, we study the following research question:

Can a schema-guided retrieval pipeline organize financial decision advice into coherent, distinct, and directionally diverse perspective clusters that preserve conditions and caveats instead of collapsing evidence into a single synthesized answer?

While our evaluation focuses on system-level viewpoint coverage, we view this as a step toward retrieval interfaces that may support more calibrated trust and reflective decision-making in future user studies.

4 System Design: The EPIC Pipeline

We call the pipeline *EPIC: Epistemic Progressive Interest Clustering*. The name reflects the system’s stepwise transformation of retrieved financial evidence into structured argument units, perspective clusters, and deliberation-facing viewpoint panels.

4.1 Pipeline Assumptions

We make pragmatic assumptions about decision-oriented advice based on the documents.

- (1) First, relevant sources contain arguments rather than only facts.
- (2) Second, those arguments can often be decomposed into a main claim, supporting reasons, conditions, and caveats.
- (3) Third, most arguments have a broad directional orientation, even if they are nuanced or conditional.
- (4) Fourth, a useful interface should present representative perspectives rather than every minor variant found in the corpus.

4.2 The Need for Light Supervision

The EPIC pipeline is not fully unsupervised, and this is intentional. Early versions that attempted to cluster raw extracted passages produced generic “it depends” clusters because conditional language dominated the semantic representation. The current version, hence, uses light supervision at the schema level. It does not require labeled training data, hand-labeled documents, or predefined final viewpoint summaries. However, it does prespecify the structural slots that organize the search space. Therefore, we claim this to be a *schema-guided argument discovery and viewpoint induction* design.

4.3 Overview

EPIC consists of ten steps: corpus cleaning, candidate document retrieval using BM25, structured argument decomposition, epistemic posture scoring, claim-centric embedding, direction-first perspective discovery, perspective summarization, final output assembly, display-time viewpoint selection, and interface output (Figure 1). We can interpret the entire pipeline as three layers:

- (1) The first layer builds the candidate evidence space by cleaning the corpus and extracting relevant documents using BM25 retrieval.
- (2) The second layer breaks the gathered evidence into fundamental units of pipeline: structured arguments which are then used to discover perspective families through decomposition, posture scoring, claim embedding, and direction-first clustering.
- (3) The last layer summarizes all perspectives (clusters), assembles the final output, and finally selects and displays the evidence on the interface.

This layered view clarifies that EPIC is not simply a clustering algorithm. It is a retrieval-to-perspective pipeline: retrieval determines which evidence is visible, decomposition determines which units can be clustered, clustering induces the raw landscape, and the interface selector decides what should be shown to users.

4.3.1 Stage 1: Corpus Cleaning. The corpus file contains the URLs of the documents, the names and types of sources, and the raw text scraped from the URLs. Table 3 explains the source type schema for the current pipeline.

For each query, EPIC loads a query-specific corpus file and performs conservative cleaning. The system checks required fields, removes empty or very short rows, normalizes text, strips obvious boilerplate, and preserves long-form source content for extraction. This step does not produce perspectives directly, but it improves the quality of subsequent stages by ensuring we start from stable source material.

4.3.2 Stage 2: BM25 Candidate Retrieval. In the next stage, EPIC tokenizes the cleaned documents, builds a BM25 index, scores every document against the query, and retains the top K candidates. Each retrieved document is assigned a BM25 score and retrieval rank. This step makes EPIC a retrieval-aware pipeline rather than a simple clustering pipeline. It simulates what evidence the system would actually collect for a user query. Without it, we would just be clustering the mini-corpus and not actually testing the real RAG settings.

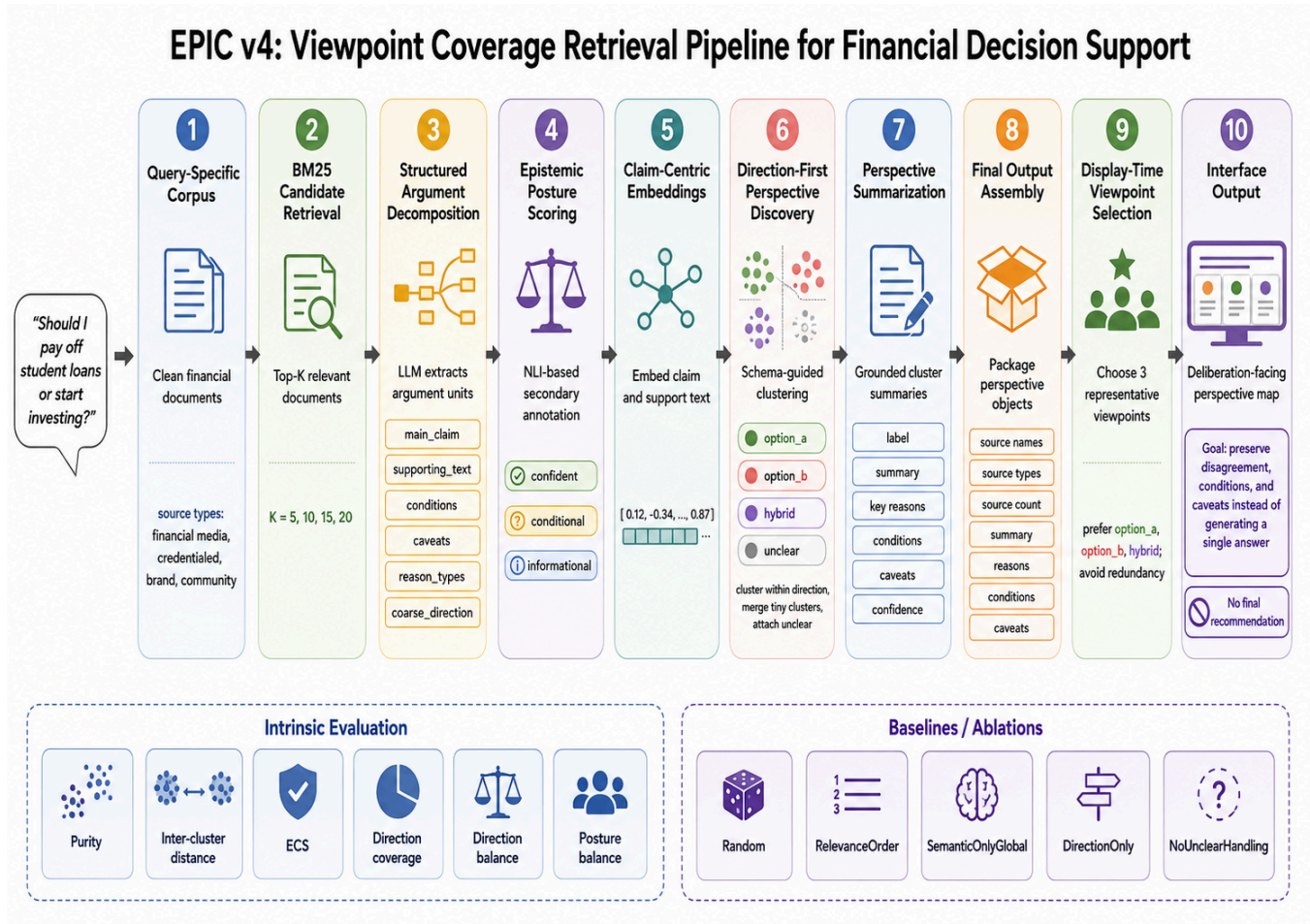


Figure 1: Overview of the EPIC v4 pipeline. EPIC retrieves query-specific financial documents, decomposes them into structured argument units, scores epistemic posture, embeds claims, discovers perspective families using direction-first clustering, summarizes clusters, and selects three representative viewpoints for a no-verdict deliberation interface.

Retrieval depth is a central design variable. It determines which viewpoints will be visible downstream, the amount of redundancy that enters the pipeline, and whether minority perspectives survive the retrieval process. A shallow pool may miss minority viewpoints, while a large pool may introduce redundant sub-variants or adjacent but mismatched evidence. We therefore evaluate EPIC at $K = 5, 10, 15, 20$ rather than treating K as a fixed engineering parameter.

4.3.3 Stage 3: Structured Argument Decomposition. Structured argument decomposition is the *conceptual backbone of EPIC*. Financial advice documents often contain multiple lines of reasoning and, sometimes, even multiple viewpoints within the same source. Therefore, treating an entire document as a single unit can obscure the different recommendations it contains. However, simply breaking documents into passages is also insufficient: early prototypes grouped full passages, but this failed because individual advice chunks often fuse together the main recommendation, reasons, thresholds, caveats, and psychological framing. If such passages are

treated as indivisible units, clustering tends to be driven by generic hedging language (generic “it depends” summaries) rather than the underlying decision direction. Splitting these monolithic argument passages allows separating what is recommended, when it applies, and what qualifies or weakens it.

EPIC therefore prompts an LLM, in our case gpt-4o-mini [3], to extract one or more structured argument units from each retrieved document. Each argument contains a `main_claim`, `supporting_text`, `conditions`, `caveats`, `reason_types`, and `coarse_direction`. For community sources, different commenter positions can become separate argument units. The `main_claim` captures the core directional thesis, while conditions and caveats are preserved separately for presentation and analysis. A few sample argument units are presented in Table 4.

4.3.4 Stage 4: Epistemic Posture Scoring. EPIC scores each structured argument for epistemic posture using an NLI-style zero-shot classifier, nli-deberta-v3-base for our pipeline (Natural Language Inference) [1, 12]. Each argument is compared against

hypotheses representing confident recommendation, conditional recommendation, and informational text. This produces a confidence score and a posture label: confident, conditional, or informational. Posture is not the primary clustering axis. Instead, it captures how strongly or cautiously an argument is made and becomes metadata for cluster characterization, evaluation, and interface display. This distinction is important because stance and posture answer different questions. Stance asks what action the argument points toward; posture asks how strongly or conditionally the argument is expressed.

4.3.5 Stage 5: Claim-Centric Embeddings. EPIC embeds both main claim and supporting_text using sentence transformer embeddings (all-MiniLM-L6-v2 [2, 20]), but uses claim embeddings as the primary representation for discovery. This is a deliberate representation choice. Raw paragraphs and supporting text are often caveat-heavy, while main claims provide a cleaner anchor for viewpoint induction. Supporting text embeddings are preserved for grounding, summarization, and future evidence retrieval, but clustering primarily operates on the claim representation.

4.3.6 Stage 6: Direction-First Perspective Discovery. The main clustering step uses direction-first perspective discovery. EPIC first partitions arguments into option_a, option_b, and hybrid buckets. It then clusters within each direction bucket using claim embeddings, attaches unclear arguments afterward to prevent them from dominating early clustering, merges tiny clusters, and retains at most a query-configured number of final clusters. (more details in Algorithm 1)

This is where EPIC differs most clearly from plain semantic clustering. A semantic-only global clusterer asks which claims are semantically similar. EPIC asks which sub-views exist within each broad stance family. This matters because financial advice often reuses the same vocabulary across opposing recommendations. Debt-first, invest-first, and hybrid arguments may all discuss interest rates, risk, goals, and returns. Without the directional scaffold, these claims can collapse into one another despite pointing toward different decisions. This step produces the raw epistemic landscape, not final summaries, not final interface cards, but the intermediate perspective families that the rest of the pipeline will utilize.

4.3.7 Stage 7: Perspective Summarization. The discovered clusters are machine-readable but not yet user-facing. EPIC therefore uses the LLM gpt-4o-mini to summarize each cluster into a short contrastive label, a concise summary, key reasons, common conditions, common caveats, and a confidence label. The summarizer is instructed to use only evidence from the cluster. This stage translates a cluster of structured argument units into a readable perspective that can be shown in the interface or discussed in the report. A few sample summaries have been presented in Table 5.

4.3.8 Stage 8: Final Output Assembly. After summarization, EPIC packages each perspective into a structured output object containing the perspective label, confidence, source types, source names, source count, summary, reasons, conditions, and caveats. This produces a clean product-style representation that can be saved, inspected, compared across runs, and fed into the Streamlit interface.

4.3.9 Stage 9: Display-Time Viewpoint Selection. The discovery step may retain five or six clusters, but an interface should not necessarily display all of them equally. We observed that raw cluster sets often contain duplicated same-side sub-variants. If the interface simply sorted clusters by confidence or size, it might display multiple variants from the same side of the decision and hide the broader landscape.

EPIC, therefore, includes a display-time selector that selects three main viewpoints for the interface. The selector prioritizes directional coverage, then confidence and semantic distinctness, while penalizing redundancy. When available, it prefers one option_a view, one option_b view, and one hybrid view. This distinction between discovery and presentation is central: Stage 6 discovers the landscape, while Stage 9 foregrounds representative viewpoints for deliberation. (more details in Algorithm 2)

4.3.10 Stage 10: Final Interface Output. The Streamlit interface presents EPIC outputs as a deliberation aid rather than an answer engine. At the top of the page, the user sees the original question and a notice explaining that experts and communities may frame the decision differently. The main view contains three perspective panels. Each panel includes a label, summary, epistemic posture, source count, key reasons, conditions, caveats, and expandable grounding evidence. A sidebar displays overview metrics, including purity, inter-cluster distance, direction balance, and posture balance. A bottom notice states that the interface does not make a recommendation. A sample interface screenshot is presented in Figure 2.

This design is intentional. If the system produces a final recommendation after presenting multiple perspectives, the recommendation could dominate user judgment and undermine the interface’s purpose. By withholding a verdict, the interface encourages users to compare conditions, inspect evidence, and retain decision authority.

5 Evaluation

5.1 Experimental Setup

We evaluate EPIC v4 across five financial decision queries (Table 1) and four retrieval depths: $K = 5, 10, 15, 20$. For every query- K pair, the pipeline retrieves candidate documents, extracts structured arguments, discovers perspective clusters, and produces three interface-level viewpoints using a display-time selector. We track the number of retrieved documents, extracted arguments, discovered clusters, and intrinsic quality metrics for each run.

5.2 Metrics

We evaluate the quality of perspective clusters using both geometric and composite epistemic metrics. It is worth noting that no single metric captures all desirable properties of a deliberation support system, so we use a family of measures that collectively capture different aspects of cluster quality. (Table 6)

- (1) Geometric Coherence and Separation:** *Purity* measures mean pairwise cosine similarity among claim embeddings within each cluster. Higher values indicate tighter, more coherent clusters. *InterDist* measures the mean cosine distance

Table 1: Financial decision queries tested in the EPIC.

Query ID	Decision question	Expected viewpoint families
q1	Should I pay off student loans or start investing?	Debt-first; invest-first; hybrid
q2	Should I contribute to a Roth IRA or a traditional 401(k)?	Roth/tax-free growth; traditional/tax deferral; mixed tax strategy
q3	Should I build an emergency fund or pay down credit-card debt first?	Emergency fund first; debt first; staged or balanced strategy
q4	Is buying a home better than renting given current interest rates?	Buy; rent; conditional long-term/short-term trade-off
q5	Should I invest in index funds or pay off my mortgage early?	Invest first; mortgage payoff first; security/return trade-off

between cluster centroids. Higher values indicate that perspective families are meaningfully separated. These two metrics are necessary but not sufficient: a system could achieve high *purity* through overly narrow clusters, or high *Inter-Dist* by collapsing minority viewpoints while keeping the remainder far apart.

- (2) **Epistemic Coverage Score (ECS):** To jointly reward coherence and separation, we introduce ECS, which is inspired by the log-scaling used in TF-IDF and similar information retrieval measures. For each cluster, ECS computes the product of its purity (local concentration, similar to TF) and a log-ratio term reflecting how distinct the cluster is from the rest of the set (analogous to how IDF rewards terms that are rare across documents, global distinctiveness). Summing across all clusters produces a composite that rewards tight, well-separated perspective families. Like TF-IDF, ECS remains a geometric measure and does not directly capture whether the cluster set is epistemically balanced or complete, which actually motivates the balance-adjusted variants below.
- (3) **Balance-adjusted scoring:** Geometric quality alone does not guarantee epistemic usefulness. A system could produce tight, well-separated clusters while still assigning 90% of arguments to one decision direction and ignoring the rest. To capture this, we introduce two balance-adjusted variants of ECS. *CovAdj* scales ECS by directional and posture coverage; it rewards systems that have at least one representative cluster for each major category (option_a, option_b, hybrid; confident, conditional, informational). *BalAdj* scales ECS by the normalized Shannon entropy [23] of those same distributions, which is stricter: it rewards not just showing up to every category but representing them proportionally. A system with ten debt-first clusters and one token hybrid cluster may score well on *CovAdj* but will be penalized by *BalAdj* (low entropy in this case, since distribution is skewed). We use *BalAdj* as our primary evaluation metric because it most directly reflects the deliberation-support goal of our system: a useful perspective map should be geometrically clean, directionally complete, and proportionally balanced across stance families.
- (4) **Entropy diagnostics:** *PostureBal* and *DirectionBal* are standalone normalized Shannon entropy scores over posture and direction distributions, bounded in [0,1]. They help diagnose why a particular run achieves its *BalAdj* score; whether low balance stems from direction skew, posture skew, or both.

5.3 Baselines and Ablations

We compare EPIC v4 against five baselines designed to isolate the contribution of each major design decision in the pipeline. Together they answer a layered question: is clustering doing anything at all, is it retrieval or clustering driving results, and does each added layer of structure: directional scaffolding, semantic sub-clustering, and unclear handling actually earn its keep.

- **RANDOM** randomly assigns arguments to clusters and serves as a lower bound. If EPIC cannot beat this, the clustering pipeline is not meaningful.
- **RELEVANCEORDER** groups arguments by BM25 retrieval rank rather than semantic or directional structure, testing whether retrieval order alone induces perspective structure.
- **SEMANTICONLYGLOBAL** applies global clustering to claim embeddings without directional scaffolding or delayed unclear handling. This is the most important intrinsic baseline. It prevents overclaiming by testing whether EPIC’s added structure helps beyond ordinary embedding clustering.
- **DIRECTIONONLY** groups arguments solely by coarse direction (option_a, option_b, hybrid) without semantic refinement within those buckets, testing whether within-direction sub-clustering contributes beyond stance partitioning alone.
- **NOUNCLEARHANDLING** uses the direction-first setup but removes delayed treatment of unclear arguments, directly testing whether allowing vague or meta-level arguments into early clustering degrades quality or balance.

Table 7 provides more details about these ablations.

5.4 Aggregate Results

Table 8 summarizes results across all queries, retrieval depths, and methods. Table 2 summarizes the strongest balance-adjusted EPIC run for each query. Two patterns are immediately visible. First, EPIC performs most strongly on Q1 and Q3, where it clearly leads on *BalAdj* at K=15 (7.84 and 8.37, respectively) and outperforms **SEMANTICONLYGLOBAL** by a substantial margin. On Q2, Q4, and Q5, the advantage is smaller or reversed, for query-specific reasons rather than reflecting a general failure of the pipeline. Second, **NOUNCLEARHANDLING** frequently achieves high raw *BalAdj* scores; sometimes exceeding EPIC, but as discussed below, this reflects numerical inflation rather than qualitatively better outputs. **RANDOM** and **RELEVANCEORDER** consistently occupy the bottom of the ranking, confirming that neither arbitrary grouping nor retrieval order alone produces meaningful perspective structure.

Table 2: Best EPIC runs by balance-adjusted score across the five-query sweep. The best numerical run is not always the best interface run; Q2 is discussed separately because its $K = 5$ result is narrow despite a higher score.

Query	Best K	EPIC bal.-adj.	Semantic comparison	Interpretation
q1	15	7.84	EPIC clearly ahead	Strong three-family landscape
q2	5	2.78	EPIC ahead numerically	Noisy corpus alignment; $K = 15$ better qualitatively
q3	15	8.37	EPIC clearly ahead	Strongest generalization case
q4	20	1.97	Semantic-only slightly ahead	Asymmetric buy-side duplication
q5	15	2.16	Semantic-only slightly ahead	Mostly two-family landscape

5.5 Retrieval Depth as a Control Knob

Retrieval depth substantially shapes the epistemic landscape available to the pipeline, and this effect is directly evident in the argument pool size.

For Q1, moving from $K=5$ to $K=15$ roughly triples the number of extracted arguments (9→32), which directly helps in the discovery of a third perspective family that is absent at shallow depths. A similar pattern holds for Q3 (10→30 args) and Q4 (10→34 args).

- (1) At $K=5$, the system typically produces only two clusters and often misses a major viewpoint family entirely.
- (2) At $K=10$, outputs are usually usable, but minority or hybrid views remain underrepresented.
- (3) $K=15$ is the strongest general-purpose setting, achieving the best *BalAdj* for Q1, Q3, and Q5 and producing *the most complete three-family perspective maps*. It often achieved the best balance between coverage and redundancy.
- (4) At $K=20$, additional documents occasionally add useful sub-variants, but more often they increase redundancy on one side of the decision space without meaningfully growing the argument pool. For example, Q1 adds only 6 arguments moving from $K=15$ to $K=20$ (32→38) while gaining no new clusters.

This finding is central to the project. In standard retrieval settings, top- K is often simply treated as an "implementation setting". *In viewpoint coverage retrieval, it directly determines which perspectives are available for discovery.* More retrieval is not always better: it increases argument volume faster than it increases the number of genuinely distinct user-facing perspectives.

5.6 Query-by-Query Results

5.6.1 Q1: Student loans versus investing. Q1 is one of the strongest results. The corpus naturally contains debt-first, invest-first, and hybrid families. The best run is $K = 15$, with 15 retrieved documents, 32 structured arguments, and six discovered clusters. EPIC achieves *BalAdj*=7.84 at this depth, compared to 6.50 for SEMANTICONLY-GLOBAL and 6.71 for RANDOM. The diagnostic table (Table 9) confirms strong directional and posture balance at $K=15$ (*DirBal*=0.958, *PostBal*=0.895), indicating that the discovered clusters are not only geometrically clean but epistemically well-distributed. The three main interface viewpoints are *High-Interest Debt First*, *Invest Over Low-Interest Loans*, and *Balanced Approach: Debt and Investment*. These outputs are highly legible and correspond to common reasoning patterns in finance advice literature: repay expensive debt

for a guaranteed return, invest when loan rates are low and tax-advantaged accounts are available, or split available cash flow when both goals matter.

5.6.2 Q2: Roth IRA versus traditional 401(k). Q2 is the weakest-aligned query. The corpus frequently discusses adjacent comparisons: Roth versus traditional accounts generally, or IRA versus 401(k), rather than the exact Roth IRA versus traditional 401(k) framing in line with the query. This corpus-query misalignment is also evident in the argument pool: even at $K=20$, only 32 structured arguments are extracted, fewer than Q1 and Q3 at the same depth, suggesting that the retrieved documents contain *fewer on-topic claims. All methods score low as a result.* The diagnostic table reveals that *DirBal* for EPIC never exceeds 0.756 across any K , while *PostBal* is consistently higher, indicating the problem is directional coverage rather than posture diversity. In other words, sources are epistemically varied in tone but not in the specific decision direction they address. Although $K=5$ gives the best numerical EPIC score (*BalAdj*=2.78), the output is too narrow. $K=15$ is the more useful setting qualitatively with 26 arguments, producing three clusters that cover the following topics: *Roth IRA for Future Tax Benefits*, *Traditional 401k for Immediate Tax Benefits*, and *Roth IRA for Tax-Free Growth*. This output is fairly readable, but skewed toward Roth-side variants and lacks a strong hybrid family. Q2, therefore, shows that viewpoint discovery quality depends not only on the clustering algorithm but also on whether the *source ecosystem supports the exact comparison requested by the user.*

It is to be noted that EPIC does not compensate for thin evidence by hallucinating perspectives or inflating cluster diversity beyond what the retrieved documents support. The output is narrow because the evidence is narrow. It is a form of graceful degradation that contrasts with LLM-based systems, which may generate fluent but unsupported viewpoints when source coverage is weak [4, 16].

5.6.3 Q3: Emergency fund versus credit-card debt. Q3 is the strongest generalization case. The corpus naturally separates into three recurring families: build a financial cushion first, pay down high-interest credit-card debt first, and do both in stages. At $K=15$, 15 retrieved documents yield 30 structured arguments and six discovered clusters (a similar pipeline depth to Q1), and EPIC achieves *BalAdj*=8.37, its highest score across all queries and depths. SEMANTICONLYGLOBAL drops sharply to 2.87 at the same setting, the largest gap between EPIC and semantic-only across all evaluations. The diagnostic table shows near-perfect directional balance at $K=20$ (*DirBal*=0.996), confirming that EPIC successfully induces all three stance families. The main viewpoints are *Emergency Fund*

Before Debt, High-Interest Debt First, and Balance Savings and Debt Repayment. The conditions attached to each perspective are also decision-useful: unstable income strengthens the emergency-fund case, high interest rates strengthen the debt-first case, and staged strategies apply when users need both liquidity and debt control. This result suggests that EPIC generalizes well when the corpus contains naturally separable stance families.

5.6.4 Q4: Buying a home versus renting. Q4 reveals a structurally asymmetric decision space. Many sources talk about nuanced reasons to buy for long-term wealth, while rent-side arguments focus on flexibility and current affordability. At $K=20$, 20 documents yield 34 arguments, which is technically a healthy pool, but *DirBal* for EPIC never exceeds 0.625 across any depth, indicating the direction distribution is skewed regardless of how much evidence is retrieved. This is possibly a corpus structure problem rather than a retrieval depth problem: more documents do not help because the available sources are themselves asymmetrically distributed across stances. SEMANTICONLYGLOBAL is slightly ahead on BALADJ at $K=20$ (2.06 versus 1.97), suggesting that for asymmetric corpora, global semantic clustering may handle imbalance better than direction-first scaffolding. The main viewpoints include *Buy for Long-Term Wealth* and *Rent for Flexibility and Cost Savings*, but the third displayed view duplicates a buy-side subtype (with different conditions and caveats). This output is meaningful, but less balanced than Q1 or Q3. It shows that broader macroeconomic decisions may produce asymmetric stance spaces, with one side having more sub-variants than the other.

5.6.5 Q5: Index funds versus mortgage payoff. Q5 is usable but narrower than Q1 or Q3. The corpus mostly centers on two recurring families: invest when mortgage rates are low, or pay off the mortgage for certainty and emotional relief. At $K=15$, EPIC achieves BALADJ=2.16 compared to 2.30 for SEMANTICONLYGLOBAL, a small gap that reflects the naturally two-family structure of this decision rather than a clear pipeline failure. Table 8 shows that $K=15$ and $K=20$ produce identical results across all methods (12 docs, 31 args, 3 clusters), confirming that the corpus reached its ceiling at 12 cleaned documents and additional retrieval depth provides no new evidence. The main viewpoints are *Invest Over Low-Rate Mortgage, Mortgage Freedom First, and Mortgage Payoff for Security*. This is a good output, but the third perspective is closer to a sub-variant of mortgage payoff than a fully separate family. Q5, therefore, suggests that some financial decisions naturally support two strong epistemic families rather than three equally distinct views.

5.7 Role of the Display-Time Selector

Several raw cluster sets contained duplicated same-side sub-variants. The display-time selector improves the interface by preferring one strong option-A view, one strong option-B view, and, when available, one hybrid view. This was especially useful for Q2, where Roth variants dominated; Q4, where buy-side clusters duplicated one another; and Q5, where mortgage-payoff variants occupied multiple slots. This supports a useful distinction between discovery and presentation. Discovery may retain five or six clusters for analysis, while the user-facing interface should foreground the three most representative and distinct viewpoints.

5.8 Summary of Findings

Taken together, the results support four primary findings:

- (1) EPIC works best as a structured argument discovery and viewpoint induction pipeline, not as a universally superior intrinsic clusterer.
- (2) The pipeline generalizes unevenly but meaningfully: it performs strongly on Q1 and Q3, moderately on Q5, and more weakly on Q2 and Q4.
- (3) Retrieval depth is a *first-class experimental variable* for epistemic retrieval.
- (4) EPIC’s strongest likely advantage is human-centered usefulness: even when semantic-only remains competitive on raw metrics, EPIC often produces outputs that are easier to present as deliberation-facing perspective maps because they preserve stance families, conditions, and caveats.

6 Discussion

6.1 What We Learned

The five-query sweep substantially changes the interpretation of the project. EPIC is no longer only a promising single-query prototype. It is a reusable framework that can be run across multiple financial decision questions, but its success depends strongly on the *structure of the query and the available source ecosystem*.

The first lesson is that viewpoint discovery for financial advice requires representation design (splitting monolithic arguments into smaller units). Clustering full passages tends to mix claims, conditions, and caveats, producing generic conditional summaries. Separating main claims from conditions and caveats made the resulting perspective maps more interpretable across multiple queries.

The second lesson is that retrieval depth is a core design variable. Very small retrieval pools can miss major viewpoint families, while larger pools can add redundancy faster than they add new perspectives. This suggests that epistemic retrieval systems should evaluate retrieval depth sensitivity rather than treating K as a fixed engineering default.

The third lesson is that corpus-query alignment is as important as clustering quality. Q2 performed poorly, not simply because the clustering failed, but because the source ecosystem often discussed nearby comparisons rather than the exact Roth IRA versus traditional 401(k) decision.

The fourth lesson is that semantic-only clustering is a stronger baseline than expected. EPIC outperforms semantic-only on some settings, especially Q1 and Q3 at mid-range retrieval depths, but not uniformly. Therefore, we do not frame contribution as “direction-aware clustering always beats semantic clustering.” *A more accurate contribution is that schema-guided decomposition and direction-aware organization produce more interpretable deliberation-facing outputs, especially when the underlying corpus contains separable stance families.*

6.2 Challenges and Design Decisions

The main technical challenge was avoiding collapse into generic “it depends” clusters. Financial advice is highly conditional, so the same passage often contains a recommendation, a threshold, a

caveat, and an exception. EPIC addresses this by breaking a monolithic argument into separate claims, conditions, and caveats before embedding and clustering.

A second challenge was generalization. Early versions used query-specific labels such as `debt_first` and `invest_first`. EPIC v4 replaces these with query-relative labels: `option_a`, `option_b`, `hybrid`, and `unclear`. This change allowed the same notebook to run across five financial decision queries while preserving a consistent evaluation structure.

A third challenge was presentation. Discovery can reasonably keep several sub-variants (even the redundant ones), but the interface becomes confusing if multiple displayed panels represent the same side of the decision. The display-time selector was thus added to separate the analytical goal of discovering clusters from the interface goal of foregrounding the most representative and distinct perspectives.

6.3 Limitations

The current study has several limitations. First, the corpora are still relatively small, and some queries have fewer than 20 cleaned documents. This limits the extent to which BM25 acts as a meaningful retrieval bottleneck. Second, structured decomposition and summarization rely on language model prompts, which may introduce extraction errors or use parametric knowledge despite grounding instructions. Third, the direction labels are query-relative but still lightly supervised, so EPIC is better described as schema-guided viewpoint discovery than fully unsupervised clustering. Fourth, our evaluation is primarily intrinsic. We have not yet collected human judgments of perspective coherence, distinctness, usefulness, or faithfulness. Fifth, the best numerical run is not always the best interface run, as shown by Q2; this indicates that intrinsic metrics alone are insufficient for judging deliberation-support systems.

6.4 Future Work

Future work should move in three directions. First, the system should be evaluated with larger, more diverse candidate corpora to give retrieval a stronger role in shaping the evidence pool. We also wish to make this system query-agnostic. Second, human annotation should be used to validate whether clusters are perceived as coherent, distinct, useful, and faithful to the underlying sources. Third, a user study should compare standard search, single-answer RAG, and EPIC-style perspective panels on trust calibration, perceived agency, caveat awareness, and decision quality. The five-query sweep also suggests a methodological direction: future systems may need query reformulation or corpus-alignment diagnostics before viewpoint induction, especially when the user’s decision framing does not match how source documents naturally organize the topic.

7 Conclusion

This paper introduced EPIC, a prototype epistemic RAG pipeline for viewpoint coverage retrieval in financial decision support. EPIC retrieves candidate documents, decomposes them into structured argument units, clusters claims using a direction-aware procedure, and presents grounded perspective panels without making a final recommendation. In a five-query sweep across four retrieval depths,

EPIC performed best on decisions whose source corpora naturally contained separable stance families, especially student-loan repayment versus investing and emergency funds versus credit-card debt. It was less successful when the corpus only weakly matched the exact query framing or when the decision space was naturally asymmetric. The results also show that retrieval depth strongly shapes the discovered epistemic landscape and that semantic-only clustering remains a strong intrinsic baseline. The broader contribution is therefore not a claim of definitive algorithmic superiority, but a lesson in task framing and system design: decision-support RAG should preserve disagreement, assumptions, and caveats rather than compressing evidence into a single synthesized answer.

8 Contributions

- **Aryan Ramchandra Kapadia:** Took the lead on the pipeline implementation, experimental evaluation, interface design, and manuscript integration. Wrote the introduction, system design, evaluation, and discussion sections.
- **Akhil Gogineni:** Worked on corpus collection, pipeline refinements, and interface wiring. Wrote related work, problem formulation, and conclusion sections.

9 AI Usage Disclosure

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A Algorithms, Tables and Figures

Algorithm 1 Direction-First Perspective Clustering

Input: structured arguments A , claim embeddings E , distance threshold τ , minimum cluster size m , maximum clusters K
Output: final cluster set C_{final}

```

1:  $C \leftarrow \emptyset$ 
2: for all  $d \in \{\text{option\_a}, \text{option\_b}, \text{hybrid}\}$  do
3:    $A_d \leftarrow \{a \in A \mid a.\text{coarse\_direction} = d\}$ 
4:   if  $|A_d| > 0$  then
5:      $C_d \leftarrow \text{AGGLOMERATIVECLUSTER}(A_d, E, \tau)$ 
6:      $C \leftarrow C \cup C_d$ 
7:   end if
8: end for
9: for all  $u \in A$  such that  $u.\text{coarse\_direction} = \text{unclear}$  do
10:   $c^* \leftarrow \text{NEARESTCENTROIDCLUSTER}(u, C)$ 
11:  assign  $u$  to  $c^*$ 
12: end for
13: while  $\exists c \in C$  such that  $|c| < m$  do
14:   $c' \leftarrow \text{NEARESTCOMPATIBLECLUSTER}(c, C)$ 
15:  merge  $c$  into  $c'$ 
16:  remove  $c$  from  $C$ 
17: end while
18:  $C \leftarrow \text{RANKCLUSTERS}(C)$ 
19:  $C_{final} \leftarrow \emptyset$ 
20: for all  $d \in \{\text{option\_a}, \text{option\_b}, \text{hybrid}\}$  do
21:   $c_d \leftarrow$  highest-ranked cluster in  $C$  with dominant direction  $d$ 
22:  if  $c_d$  exists then
23:     $C_{final} \leftarrow C_{final} \cup \{c_d\}$ 
24:    remove  $c_d$  from  $C$ 
25:  end if
26: end for
27: while  $|C_{final}| < K$  and  $C \neq \emptyset$  do
28:  add the highest-ranked cluster in  $C$  to  $C_{final}$ 
29:  remove that cluster from  $C$ 
30: end while
31: return  $C_{final}$ 

```

Algorithm 2 Main Viewpoint Selection for Interface Display

Require: Cluster set C , display budget K

Ensure: Display viewpoint set $\mathcal{V}$

```

1:  $\mathcal{V} \leftarrow \emptyset$ 
2: for each direction  $d \in \{\text{option\_a}, \text{option\_b}, \text{hybrid}\}$  do
3:   select the highest-ranked cluster in  $C$  with dominant direction  $d$ , if one exists
4:   add it to  $\mathcal{V}$  if not already selected
5:   if  $|\mathcal{V}| = K$  then
6:     return  $\mathcal{V}$ 
7:   end if
8: end for
9: while  $|\mathcal{V}| < K$  do
10:  select the remaining cluster that maximizes confidence and semantic distinctness from clusters already in  $\mathcal{V}$ 
11:  add it to  $\mathcal{V}$ 
12: end while
13: return  $\mathcal{V}$ 

```

Table 3: Source type schema used in EPIC v4.

Source type	Definition	Examples
brand	Content published by a company, product, bank, lender, brokerage, insurer, or platform speaking in its institutional voice. These sources may be informational or advisory, but are not primarily independent journalism or individual expert commentary.	SoFi; ELFI; Empower; T. Rowe Price; PNC; Navy Federal
financial_media	Editorial or journalistic sources that publish reported articles, explainers, and consumer-finance guidance for a broad audience. These are typically independent media outlets rather than firms promoting their own financial products.	CNBC; CBS News; Investopedia; Today's Homeowner
credentialed	Content authored by identifiable experts or professional organizations with domain-relevant expertise, credentials, or advisory authority. This includes individual financial professionals, physician-finance specialists, or expert-led advisory publications.	White Coat Investor; Advisorpedia; Vision Retirement
community	User-generated discussion spaces where advice is expressed through personal experience, peer recommendation, or informal debate rather than institutional or editorial authorship. These sources often surface minority views, practical heuristics, and psychologically salient concerns.	Reddit; Bogleheads forum; community discussion threads

Q	Source	Main claim	Conditions	Caveats
q1	White Coat Investor (<i>credentialed</i>)	You should prioritize paying off high-interest debt before investing.	If your debt is at 8% or higher; if you have credit card debt accumulating interest.	Not all debts are equal; consider the interest rates. Ensure you are not neglecting tax-advantaged investment opportunities.
q2	Fidelity (<i>brand</i>)	You should consider contributing to a traditional 401(k) to take advantage of employer contributions and higher contribution limits.	Your employer offers a 401(k) plan with matching contributions; you want to maximize retirement savings with higher contribution limits.	You will have to pay taxes on withdrawals in retirement. You may be required to take required minimum distributions (RMDs) starting at age 73.
q3	UMB Blog (<i>brand</i>)	Paying down high-interest credit card debt should be prioritized to save money in the long run.	You have high-interest credit card debt; you can make larger payments toward your debt.	Focusing solely on debt repayment may deplete your savings. It may be easier to pay off lower-balance cards for psychological benefits.
q4	Today's Homeowner (<i>financial_media</i>)	In some areas, long-term renting is cheaper than homeownership, especially given the current high interest rates and home prices.	High interest rates; soaring home prices.	Homeownership may still be beneficial in some markets. Homeowners may save in the long term if they stay in their homes.
q5	Homestead (<i>brand</i>)	Financially speaking, it is best to invest in index funds rather than pay off your mortgage early.	You must have a stable source of income that allows you to invest in a long-term strategy; you should be able to ride out tough times and keep your money invested.	Paying off your mortgage early may provide peace of mind. Real-life financial situations may vary, and personal circumstances should be considered.

Table 4: Sample structured arguments extracted by EPIC v4 across the five decision queries. Each argument is decomposed into a main claim, conditions under which it holds, and caveats that qualify or limit the recommendation.

Perspective label	Summary	Typical conditions / caveats
High-Interest Debt First	This perspective prioritizes paying off student loans, especially when interest rates are high, before investing. The core reasoning is that debt repayment offers a guaranteed return and reduces financial burden, making it preferable to uncertain market gains. It is especially appealing to users with lower risk tolerance or a strong aversion to debt.	<i>Conditions:</i> debt at 8% or higher; strong aversion to debt. <i>Caveats:</i> do not neglect tax-advantaged investment opportunities; financial situations and risk tolerance may change.
Invest Over Low-Interest Loans	This perspective favors investing over accelerated loan repayment when student loan rates are relatively low. The central idea is that long-run investment returns, especially in tax-advantaged accounts, may exceed the savings from early debt repayment. This view is often paired with arguments about employer retirement matches and the opportunity cost of delaying investing.	<i>Conditions:</i> student loan interest rate below 6%; access to tax-sheltered accounts such as a 401(k) or Roth IRA. <i>Caveats:</i> investing carries market risk; psychological benefits of becoming debt-free may still matter.
Balanced Approach: Debt and Investment	This perspective recommends splitting available resources between debt repayment and investing rather than choosing only one path. It frames the problem as a tradeoff that can often be managed simultaneously, particularly when the user has enough disposable income to make progress on both fronts. It also emphasizes securing employer matching contributions before making more aggressive tradeoffs.	<i>Conditions:</i> disposable income is available; uncertainty about which option to prioritize. <i>Caveats:</i> this approach may not maximize either outcome; progress on both fronts must still be meaningful.

Table 5: Example interface-facing perspective summaries produced by EPIC v4 for the student-loan versus investing query.

EPIC viewer

Select a run

EPIC v4 - q1_topk15

Run directory:

```
/Users/aryankapadia/Desktop/Samp/Epic_v4_topk15
```

Version: EPIC v4

Evaluation snapshot

Purity: 0.66

Inter-cluster distance: 0.29

Balance-adjusted ECS: 7.84

Posture balance: 0.90

Direction balance: 0.96

Should I pay off student loans or start investing?

Experts and communities may frame this decision differently. The views below are grounded in retrieved documents, preserve caveats and conditions, and do not force a single final answer.

Primary viewpoints shown: 3
Discovered clusters: 6
Structured arguments: 32
Brand: 4 Community: 5 Credentialed: 8

HIGH-INTEREST DEBT FIRST

The central perspective advocates for prioritizing the repayment of student loans, especially those with high interest rates, before investing. This approach is seen as a guaranteed investment, as it reduces the burden of debt and can be more beneficial than uncertain investment returns. Additionally, it aligns with a conservative financial strategy, particularly for those with a low risk tolerance or a strong aversion to debt.

Epistemic posture: Conditional 3 grounded sources

Why this view appears

- Loan rate vs expected return

Supported by: "If you have student loans with an interest rate higher than 6%, it may be more beneficial to pay off those loans rather than invest." — [Nerdwallet](#) · Financial media

- Peace of mind / debt stress

Supported by: "If you dislike debt, you may prefer to pay it off rather than invest." — [Whitecoatinvestor](#) · Credentialed

- Risk tolerance

Supported by: "You should consider your personal risk profile when deciding between paying off loans and investing." — [Whitecoatinvestor](#) · Credentialed

Common conditions

- If your debt is at 8% or higher.
- If you are close to retirement.

▼ Caveats and exceptions

- Ensure you are not neglecting tax-advantaged investment opportunities.
- Your financial situation may change, affecting your risk tolerance.

▼ Grounding and evidence

Source mix

Credentialed Financial media

Original documents

- [Whitecoatinvestor](#) · Credentialed
- [Bankrate](#) · Financial media
- [Nerdwallet](#) · Financial media
- [Whitecoatinvestor](#) · Credentialed

Supporting evidence

- [Whitecoatinvestor](#) · Credentialed

Claim: You should consider your personal risk profile when deciding between paying off loans and investing. (posture: Confident, retrieval rank: 2)

- [Bankrate](#) · Financial media

Claim: You should prioritize paying off student loans before focusing on investing. (posture: Confident, retrieval rank: 11)

- [Whitecoatinvestor](#) · Credentialed

Claim: You should prioritize paying off high-interest debt before investing. (posture: Confident, retrieval rank: 2)

INVEST OVER LOW-INTEREST LOANS

The central perspective suggests prioritizing investing over paying off low-interest student loans. This approach is based on the potential for higher returns from investments compared to the savings from loan repayment, especially when interest rates are below 6%. Tax advantages and employer retirement matches further support this strategy.

Epistemic posture: Conditional 10 grounded sources

Why this view appears

- Long-run investment upside

Supported by: "If your long-term financial goals include building savings or retiring early, consider prioritizing investments over paying off student loans early." — [Huntingtonbank](#) · Brand

- Loan rate vs expected return

Supported by: "If your loans have an interest rate below 6%, it may make sense to put more of your money towards investing." — [Johnharcock](#) · Brand

- Tax treatment

Supported by: "Invest the rest of your spare cash instead of paying down student loans faster than necessary." — [Reddit_bogleheads](#) · Community

Common conditions

- If your student loan interest rate is below 6%.
- If you have access to tax-sheltered accounts like a 401(k) or Roth IRA.

▼ Caveats and exceptions

- Investing carries risks and is not guaranteed to yield higher returns.
- Consider your current tax bracket and how it affects your decision.

▼ Grounding and evidence

Source mix

Brand Community Credentialed Financial media

Original documents

- [Reddit_bogleheads](#) · Community
- [Cnbc](#) · Financial media
- [Hatcheditorial](#) · Financial media
- [Nerdwallet](#) · Financial media
- [Huntingtonbank](#) · Brand
- [Investopedia](#) · Financial media

Supporting evidence

- [Cnbc](#) · Financial media

Claim: You should consider the potential earnings from investing compared to the savings from paying off student loans. (posture: Confident, retrieval rank: 7)

- [Reddit](#) · Community

Claim: If you are not struggling for cash flow, you should continue making minimum payments and invest. (posture: Confident, retrieval rank: 4)

- [Huntingtonbank](#) · Brand

Claim: If your long-term financial goals include building savings or retiring early, consider prioritizing investments over paying off student loans early. (posture: Confident, retrieval rank: 13)

BALANCED APPROACH: DEBT AND INVESTMENT

When uncertain about whether to pay off student loans or start investing, consider splitting your income between both options. This approach allows you to make progress on both fronts, though it may not maximize either outcome. Prioritize obtaining any employer match before deciding on further financial actions.

Epistemic posture: Conditional 1 grounded sources

Why this view appears

- Peace of mind / debt stress

Supported by: "If unsure, consider splitting your income between paying off loans and investing." — [Whitecoatinvestor](#) · Credentialed

- Employer match and compounding

Supported by: "You should prioritize getting any employer match before deciding whether to pay off debt or invest." — [Whitecoatinvestor](#) · Credentialed

Common conditions

- If you have disposable income available.
- If you are uncertain about your financial priorities.

▼ Caveats and exceptions

- This approach may not maximize either outcome.
- Ensure you are still making progress on both fronts.

▼ Grounding and evidence

Source mix

Credentialed

Original documents

- [Whitecoatinvestor](#) · Credentialed
- [Whitecoatinvestor](#) · Credentialed

Supporting evidence

- [Whitecoatinvestor](#) · Credentialed

Claim: You should prioritize getting any employer match before deciding whether to pay off debt or invest. (posture: Confident, retrieval rank: 8)

- [Whitecoatinvestor](#) · Credentialed

Claim: If you are paralyzed from doing either paying off debt or investing, consider splitting your extra money between both. (posture: Confident, retrieval rank: 8)

- [Whitecoatinvestor](#) · Credentialed

Claim: If unsure, consider splitting your income between paying off loans and investing. (posture: Conditional, retrieval rank: 2)

› All discovered perspectives and run details

This interface does not make a recommendation. It foregrounds the main perspectives in the retrieved evidence, while preserving conditions, caveats, and links back to original sources.

Figure 2: EPIC interface output for the query “Should I pay off student loans or start investing?” The interface presents three grounded perspective panels, including key reasons, conditions, caveats, and source evidence, while explicitly withholding a final recommendation.

Metric	Formula	Meaning	Limitation	Definition
Purity	$\frac{1}{ C } \sum_{c \in C} \frac{1}{ P(c) } \sum_{(i,j) \in P(c)} \cos(e_i, e_j)$	Average within-cluster semantic coherence. Higher is better.	May reward overly narrow clusters; does not capture coverage or usefulness.	Mean pairwise cosine similarity among claim embeddings within each cluster, averaged across clusters.
InterDist	$\frac{1}{ Q(C) } \sum_{(c,c') \in Q(C)} (1 - \cos(\mu_c, \mu_{c'}))$	Average distance between cluster centroids. Higher is better.	Can be high even when the system ignores minority views.	Mean cosine distance between cluster centroids.
ECS	$\sum_{c \in C} \text{Purity}(c) \log\left(\frac{ C }{\text{sim}(c, C \setminus c) + \epsilon}\right)$	Composite score rewarding coherent and separated clusters.	Geometry-based intrinsic metric; does not capture epistemic balance directly.	Inspired by TF-IDF log-scaling; purity acts as a local coherence term (analogous to TF); the log-ratio rewards global distinctiveness across the cluster set (analogous to IDF).
CovAdj	$\text{ECS} \cdot \left(\frac{\text{DirCov} + \text{PostCov}}{2}\right)$	ECS adjusted by directional and posture coverage.	Coverage alone does not ensure balanced representation.	Scales ECS by average directional and posture coverage; rewards cluster sets that include at least one representative from each major direction and posture category.
BalAdj	$\text{ECS} \cdot \left(\frac{\text{DirBal} + \text{PostBal}}{2}\right)$	ECS adjusted by directional and posture balance.	Depends on the chosen direction/posture schema.	Scales ECS by normalized Shannon entropy over direction and posture distributions; rewards proportional representation across stance families, not merely coverage. Primary metric.
PostureBal	$-\frac{\sum_{p \in \mathcal{P}} \hat{p} \log \hat{p}}{\log \mathcal{P} }$	Entropy-based balance over posture categories. Higher is better.	Sensitive to posture taxonomy.	Normalized Shannon entropy over confident, conditional, and informational posture labels.
DirectionBal	$-\frac{\sum_{d \in \mathcal{D}} \hat{d} \log \hat{d}}{\log \mathcal{D} }$	Entropy-based balance over stance directions. Higher is better.	Does not capture within-direction nuance.	Normalized Shannon entropy over option_a, option_b, and hybrid direction labels.

Table 6: Intrinsic evaluation metrics used in EPIC v4. Symbols: e_i = embedding of argument i ; μ_c = centroid of cluster c ; $P(c)$ = set of within-cluster argument pairs for cluster c ; $Q(C)$ = set of all cluster-centroid pairs; $\text{sim}(c, C \setminus c)$ = mean cosine similarity between cluster c and all other clusters; ϵ = small smoothing constant to prevent division by zero; $\hat{p}, \hat{d}$ = empirical proportions of posture and direction labels respectively; $\mathcal{P}$ = posture label set {confident, conditional, informational}; $\mathcal{D}$ = direction label set {option_a, option_b, hybrid}. Purity and InterDist measure geometric cluster quality; ECS combines them into a composite score inspired by TF-IDF log-scaling; CovAdj and BalAdj extend ECS to reward epistemic completeness and balance; PostureBal and DirectionBal provide fine-grained entropy diagnostics. BalAdj is the primary evaluation metric.

Ablation	Definition	What it tests	Why it is included
EPIC_v4	Full pipeline with structured argument decomposition, query-relative directional scaffolding, direction-first clustering, delayed handling of unclear, and cluster cleanup.	Whether the full system produces coherent, distinct, and balanced perspective clusters.	Main method under evaluation.
Random	Arguments are randomly partitioned into the same number of clusters as EPIC.	Whether EPIC improves over arbitrary grouping.	Weak sanity-check baseline; if EPIC cannot beat this, the clustering pipeline is not meaningful.
RelevanceOrder	Arguments are grouped in retrieval order rather than by semantic or directional structure. We take the full list of arguments ordered by that inherited retrieval rank and then split them into equal-sized groups.	Whether EPIC improves over the topical structure already induced by BM25 ranking.	Tests whether gains come from actual perspective discovery rather than from retrieval order alone.
SemanticOnlyGlobal	All arguments are clustered together in one global semantic clustering pass, without directional scaffolding or delayed unclear handling.	Whether EPIC’s direction-first structure improves over a strong semantic-only baseline.	Most important intrinsic baseline; prevents overclaiming by testing whether added structure helps beyond ordinary embedding clustering.
DirectionOnly	Arguments are grouped only by option_a, option_b, and hybrid, without semantic refinement inside those buckets.	Whether the directional scaffold alone explains EPIC’s performance.	Tests whether within-direction semantic clustering contributes beyond coarse stance partitioning.
NoUnclearHandling	Uses the direction-first setup but removes the special delayed treatment of unclear arguments.	Whether allowing vague or meta-level arguments to participate in clustering from the start degrades cluster quality or balance.	Directly tests one of EPIC’s design choices for avoiding collapse into generic, weakly directional clusters.

Table 7: Ablation baselines used to evaluate EPIC v4. Together, these baselines test whether EPIC’s gains come from structured decomposition, direction-aware organization, and special handling of weakly directional arguments, rather than from trivial grouping effects.

Query	K	Docs	Args	EPIC			SEMONLY			DIRONLY			NOUNCLEAR			REORDER			RANDOM		
				Pur	IDst	BalAdj	Pur	IDst	BalAdj	Pur	IDst	BalAdj	Pur	IDst	BalAdj	Pur	IDst	BalAdj	Pur	IDst	BalAdj
Q1	5	5	9	0.52	0.18	0.71	0.59	0.30	0.94	0.51	0.28	1.65	0.64	0.35	3.59	0.53	0.20	0.75	0.52	0.17	0.69
	10	10	20	0.58	0.24	3.71	0.63	0.31	4.27	0.56	0.17	2.07	0.66	0.29	6.20	0.54	0.15	3.18	0.53	0.14	3.16
	15	15	32	0.66	0.29	7.84	0.65	0.33	6.02	0.58	0.11	1.95	0.65	0.23	7.15	0.59	0.16	6.48	0.58	0.14	6.22
Q2	5	5	14	0.58	0.33	2.78	0.55	0.21	0.69	0.53	0.30	0.75	0.55	0.29	2.55	0.53	0.26	2.39	0.53	0.25	2.41
	10	10	17	0.57	0.33	0.99	0.62	0.19	0.79	0.72	0.28	2.40	0.80	0.39	4.75	0.56	0.11	0.71	0.55	0.08	0.67
	15	15	26	0.65	0.25	2.10	0.61	0.29	1.78	0.63	0.21	1.93	0.63	0.26	4.60	0.55	0.11	1.55	0.53	0.09	1.46
Q3	5	5	10	0.61	0.24	1.10	0.57	0.34	1.19	0.58	0.35	2.48	0.58	0.35	2.48	0.56	0.12	0.86	0.57	0.14	0.90
	10	10	19	0.68	0.22	4.30	0.68	0.31	4.59	0.63	0.18	2.40	0.68	0.22	4.30	0.60	0.12	3.54	0.59	0.10	3.40
	15	15	30	0.70	0.25	8.37	0.70	0.29	2.87	0.64	0.17	2.37	0.70	0.25	8.37	0.59	0.12	6.47	0.60	0.13	6.60
Q4	5	5	10	0.72	0.27	0.90	0.73	0.31	0.97	0.69	0.23	0.82	0.82	0.31	2.23	0.62	0.07	0.59	0.64	0.11	0.64
	10	10	18	0.71	0.27	0.98	0.71	0.27	0.99	0.71	0.27	0.98	0.81	0.29	2.41	0.61	0.06	0.64	0.61	0.04	0.61
	15	15	27	0.68	0.27	1.96	0.74	0.23	1.86	0.68	0.25	0.91	0.79	0.28	2.30	0.60	0.06	1.41	0.60	0.07	1.43
Q5	5	5	11	0.70	0.26	1.14	0.62	0.00	10.48	0.81	0.31	2.93	0.81	0.31	2.93	0.61	0.06	0.74	0.63	0.11	0.83
	10	10	27	0.63	0.27	2.02	0.64	0.28	2.07	0.75	0.25	2.35	0.69	0.28	5.07	0.57	0.09	1.54	0.57	0.10	1.55
	15	12	31	0.63	0.24	2.16	0.65	0.28	2.30	0.74	0.24	2.55	0.70	0.27	5.55	0.57	0.06	1.64	0.58	0.07	1.69
Q5	20	12	31	0.63	0.24	2.16	0.65	0.28	2.30	0.74	0.24	2.55	0.70	0.27	5.55	0.57	0.06	1.64	0.58	0.07	1.69

Table 8: Full evaluation results across all queries, retrieval depths (K), and methods. Docs = number of documents actually retrieved; Args = number of structured arguments extracted. These are shared across all methods since retrieval and decomposition occur before any method diverges. Columns show Purity (Pur), Inter-cluster Distance (IDst), and Balance-Adjusted ECS (BalAdj) for each method. Bold values indicate the best score for that metric at that query- K combination. EPIC = full pipeline; SEMONLY = SemanticOnlyGlobal; DIRONLY = DirectionOnly; NOUNCLEAR = NoUnclearHandling; REORDER = RelevanceOrder; RANDOM = random assignment.

Query	K	BalAdj	CovAdj	DirBal	PostBal	Clusters
Q1 loans vs. invest	5	0.706	0.771	0.946	0.579	2
	10	3.708	3.863	0.983	0.937	4
	15	7.835	8.458	0.958	0.895	6
	20	7.557	8.244	0.934	0.899	6
Q2 Roth vs. 401k	5	2.783	3.448	0.533	0.812	4
	10	0.989	1.254	0.708	0.870	2
	15	2.098	2.742	0.743	0.788	3
	20	2.100	2.543	0.756	0.896	3
Q3 emergency vs. debt	5	1.098	1.171	0.937	0.937	2
	10	4.300	4.423	0.962	0.983	4
	15	8.372	8.767	0.979	0.930	6
	20	8.003	8.448	0.996	0.899	6
Q4 buy vs. rent	5	0.900	0.970	0.625	0.613	2
	10	0.978	1.190	0.617	0.753	2
	15	1.955	2.394	0.620	0.741	3
	20	1.970	2.340	0.620	0.782	3
Q5 index vs. mortgage	5	1.141	1.397	0.851	0.783	2
	10	2.022	2.689	0.763	0.741	3
	15	2.157	2.621	0.752	0.895	3
	20	2.157	2.621	0.752	0.895	3

Table 9: EPIC v4 diagnostic metrics across queries and retrieval depths K . BalAdj is the primary metric; CovAdj, DirBal, and PostBal are diagnostics that reveal whether low balance stems from direction skew, posture skew, or both. Bold values indicate the best K for that metric within each query.

B Online Resources

Interactive demo: [EPIC Streamlit Demo](#)

Code repository: [GitHub Repository](#)